
Introduction



Course Overview

Introduction

- ▶ Mobile Programming (AC3040)
- ▶ Structure:
 - ▶ Assiduity: Lectures + Assignments
 - ▶ Midterm project
 - ▶ Final exam
- ▶ Tools:
 - ▶ NodeJS
 - ▶ A good ~~text~~code editor:
 - ▶ VS Code



Course Objectives

- ▶ At the end of this class, you will be able to:
 - ▶ Design and implement cross-platform mobile GUI
 - ▶ Customize the style of GUI elements
 - ▶ Handle and response to user activities
 - ▶ Control the navigation between screens
 - ▶ Implement the two-way communication with a server
- ▶ And beyond:
 - ▶ Understand the mobile programming model and apply this to your designs
 - ▶ Speak the mobile programming lingo
 - ▶ Have fun with mobile development, and enjoy creating apps!

Mobile Platforms

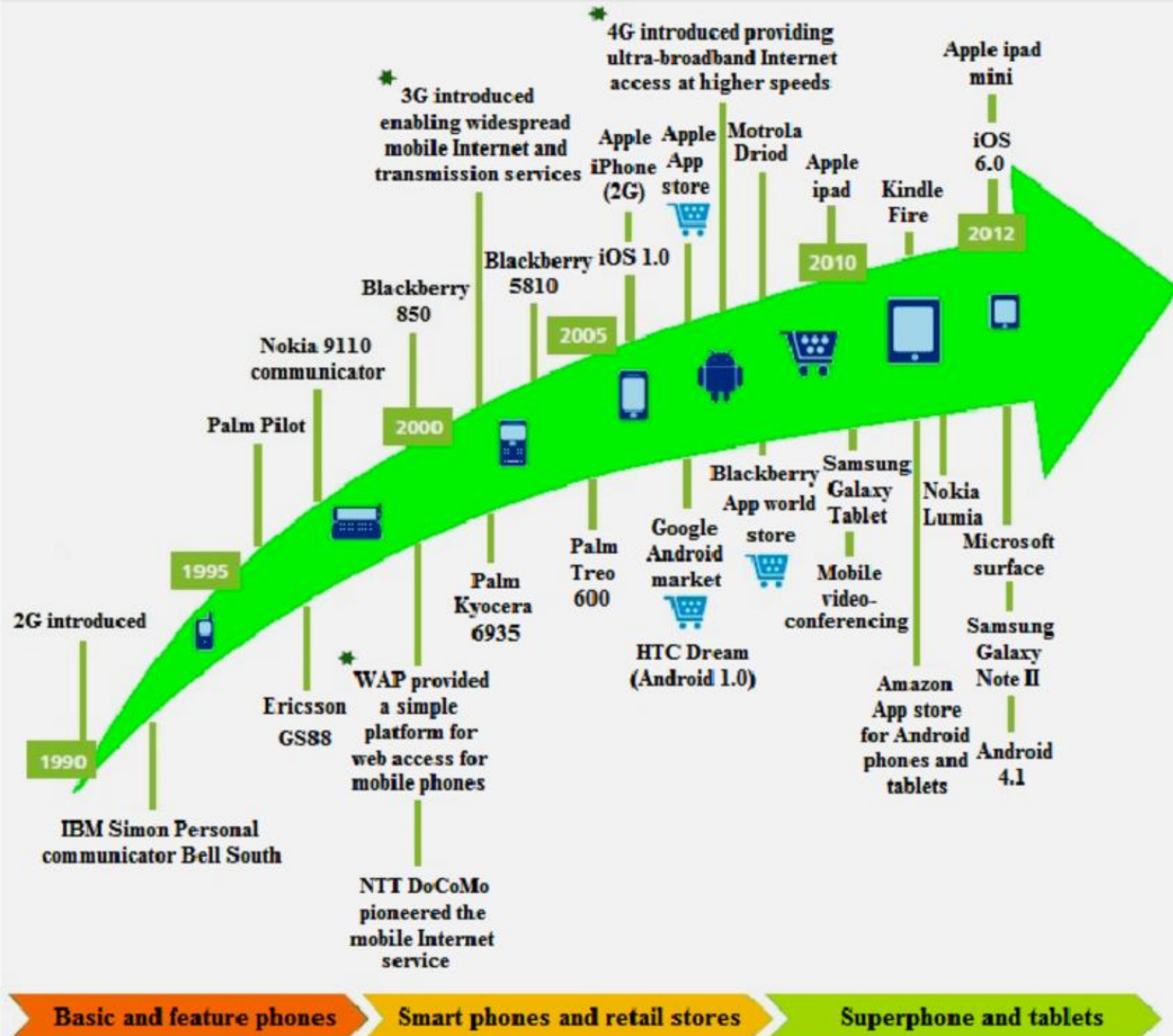
Mobile Devices

► Computers small enough to hold and operate in the hand

- Cell phones
- Laptops
- Tablets
- Smart watches
- Smart cards
- Mobile cameras
- Mobile sensors
- Robots
- Smart cars
- ...



History of Mobile Devices



Modern Mobile Devices

- ▶ Rich in types: smartphones, tablets, smart watches, smart TVs, smart cars,...
- ▶ Most common characteristics:
 - ▶ (Touchable) flat screen(s)
 - ▶ Voice capabilities
 - ▶ Wireless communications
 - ▶ Rich sensors: GPS, camera,...
 - ▶ Operating system
- ▶ Platform convergence
 - ▶ iOS
 - ▶ Android

Mobile Development

What is It?

- ▶ **Building** and **maintenance** of apps that run on mobile devices
 - ▶ Make apps work
 - ▶ Make apps work fast
 - ▶ Make apps look great
- ▶ What do mobile devs need?
 - ▶ It depends on their choices!
 - ▶ For students of this course: JavaScript

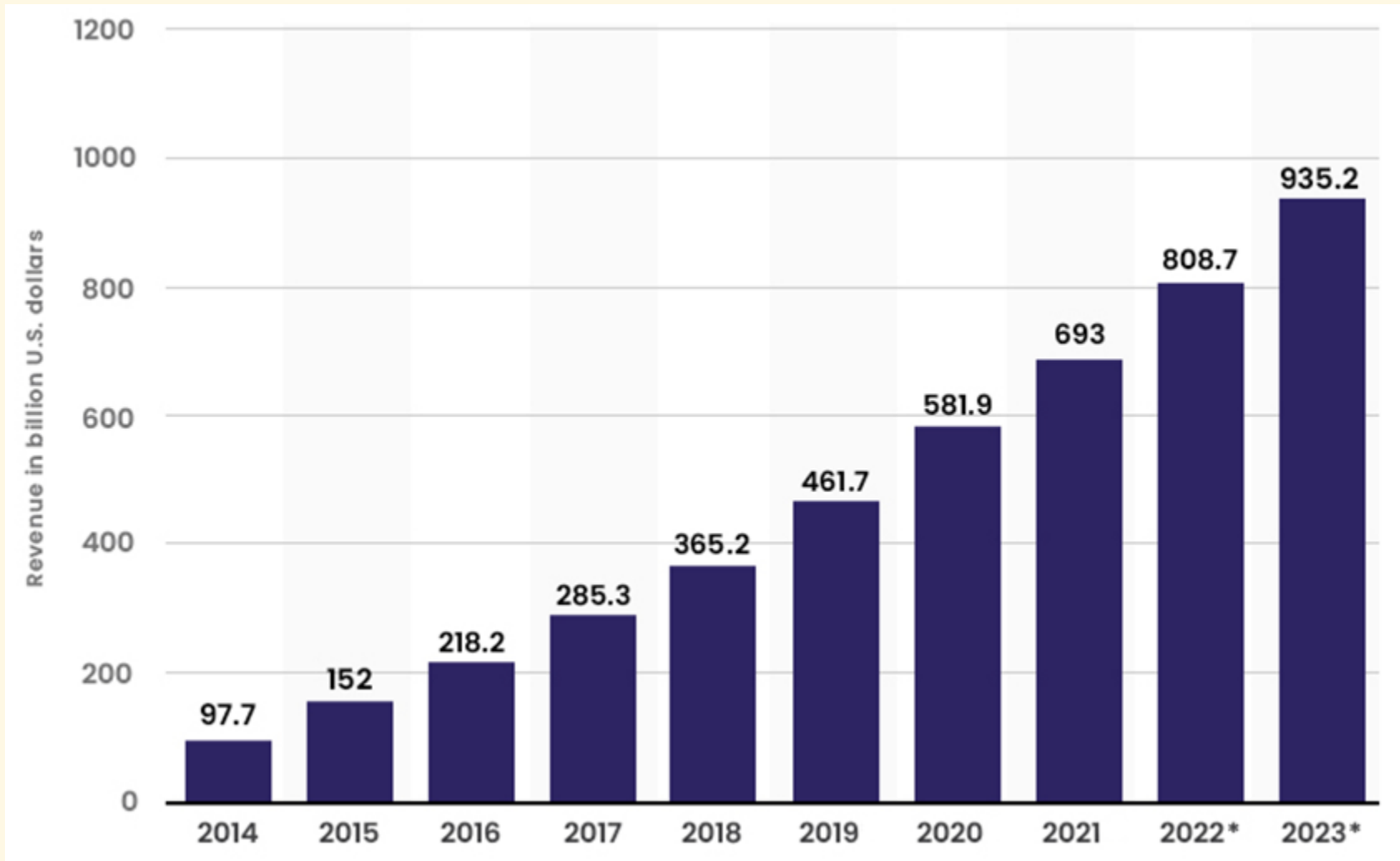


The Rise of Mobile Apps

- ▶ Internet is more and more ubiquitous, with very high-speed and wireless connections
- ▶ Mobile devices are equipped with more and more powerful capabilities
- ➔ Mobile is now one of important application/service development and distribution platforms (besides desktop, web)



Growth of Mobile App Market



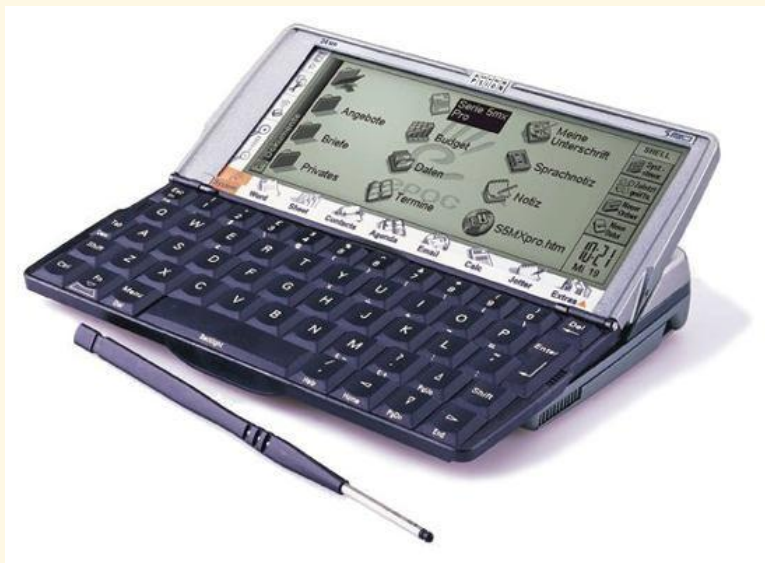
Early Mobile Development

- ▶ Low level coding
- ▶ Highly device-dependent coding
- ▶ Low availability of resources: CPU, memory
- ▶ No real OS
- ▶ 1st “smart” phone:
 - ▶ Simon in 1993 by IBM
 - ▶ Touchscreen
 - ▶ Apps: calendar, clock, email, games, notepad

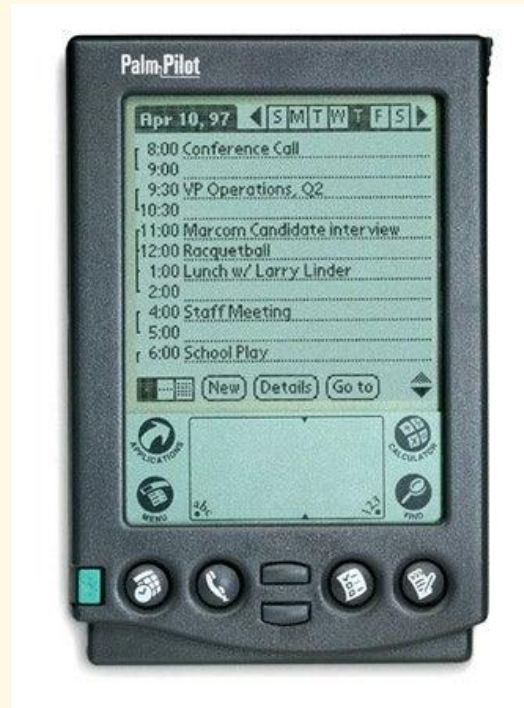


Next Mobile Platforms

- ▶ Psion EPOC in 1990s for PDAs: 32-bit OS, 2MB RAM
- ▶ Palm OS in 1996 for PDAs
- ▶ WML (Wireless Markup Language): light version of HTML
- ▶ Java ME
- ▶ Symbian



Psion EPOC



Psion EPOC



Symbian

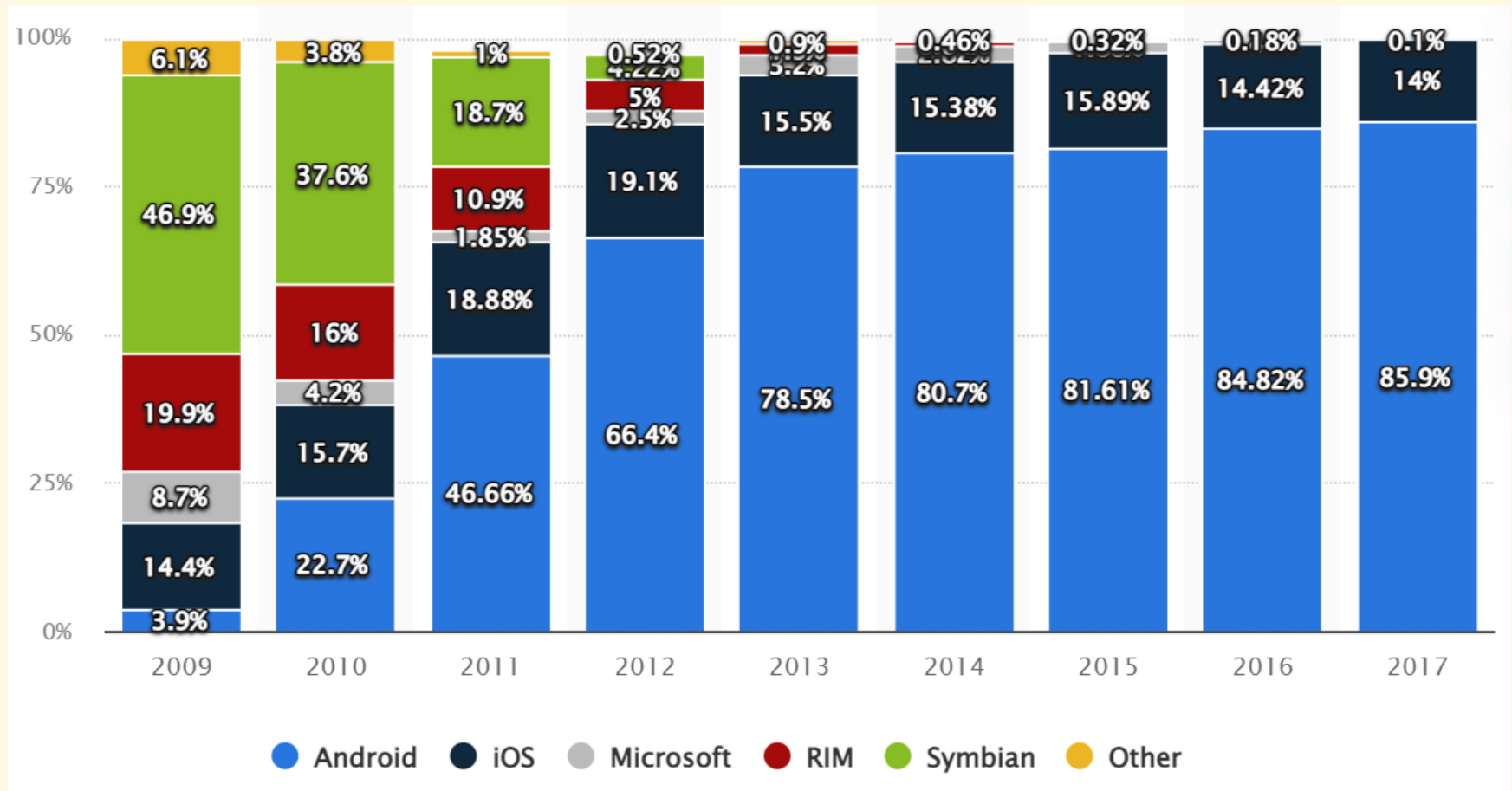
Modern Mobile Platforms

- ▶ iOS
 - ▶ Apple, 2007
 - ▶ Kernel: XNU
- ▶ Android:
 - ▶ Google, 2008
 - ▶ Kernel: Linux
- ▶ Windows Phone
 - ▶ Microsoft, 2010
 - ▶ Discontinued in 2020
- ▶ High level languages
- ▶ Fast development

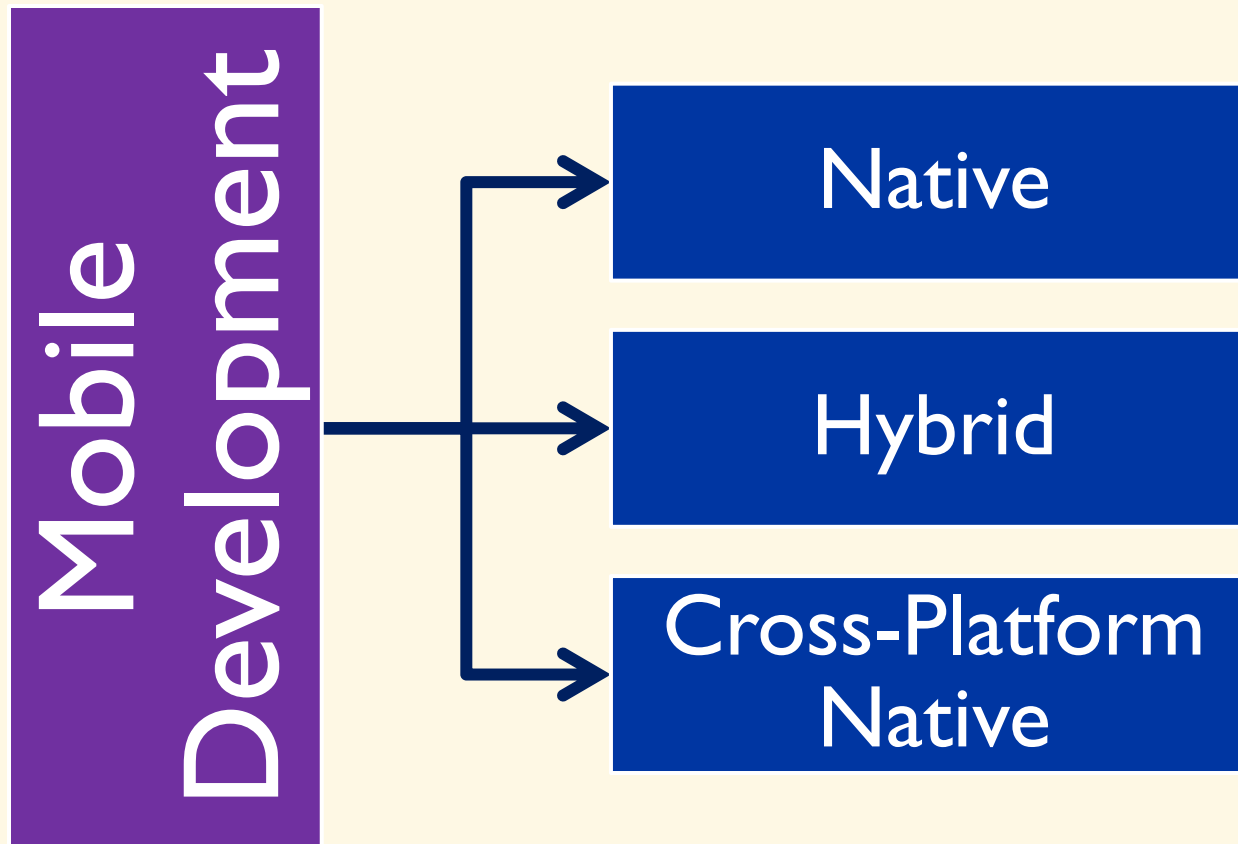


Most Common Mobile Platforms Nowadays

► Share of sales to end-users



Approach Classification



Native Approach

- ▶ Frameworks offered by the platform providers themselves
 - ▶ Best performance, compatibility, user experience
 - ▶ Not portable to other platforms
 - ▶ High maintenance cost
- ▶ Android: Java, Kotlin
- ▶ iOS: Swift, Objective-C

Hybrid Approach

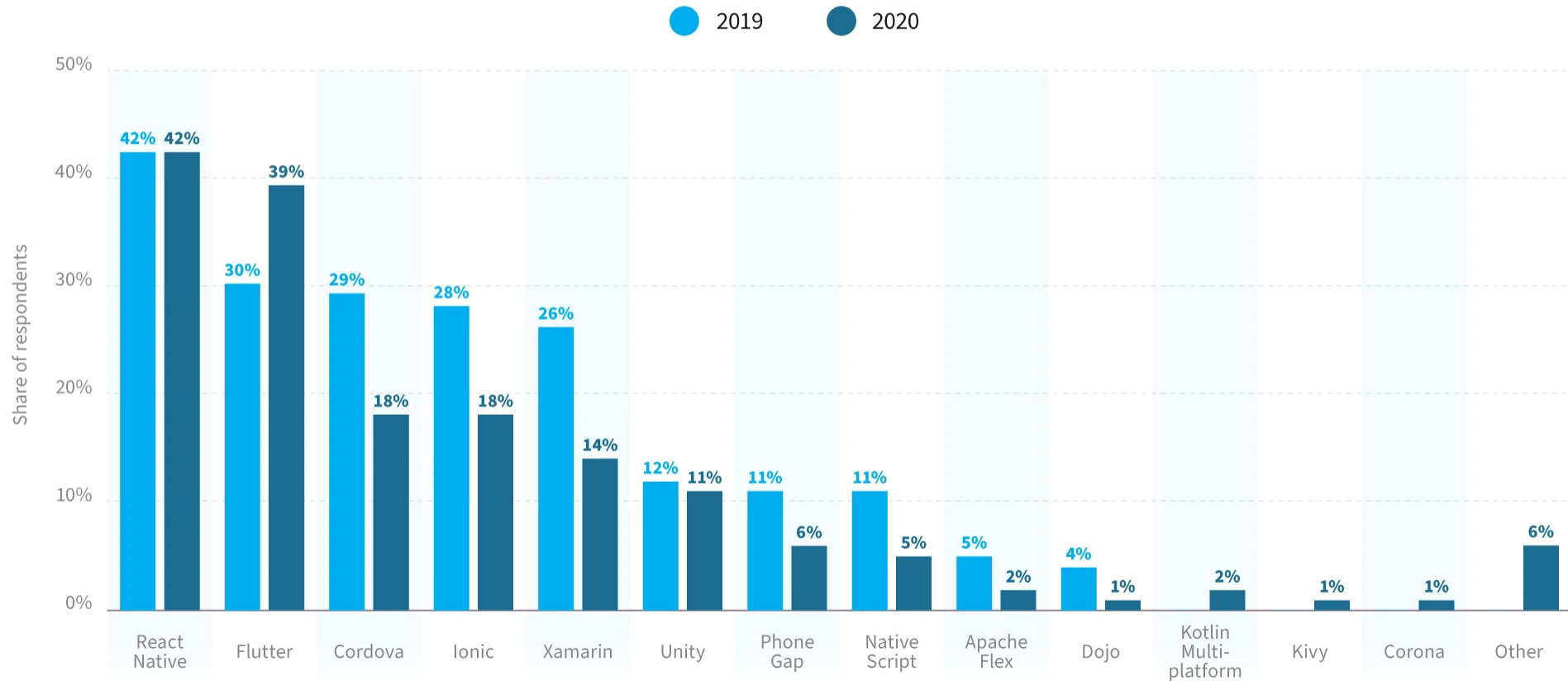
- ▶ Cross-platform frameworks using web technologies
 - ▶ HTML, CSS, JavaScript
 - ▶ App is basically composed of web pages
 - ▶ Low development cost
- ▶ Most popular ones:
 - ▶ Cordova: by Apache
 - ▶ Ionic: by Drifty

Cross-Platform Native Approach

- ▶ Cross-platform frameworks for native apps
 - ▶ Write once in an intermediate language
 - ▶ Build specifically for different platforms
 - ▶ Native UI, good performance and UX
- ▶ Most popular ones:
 - ▶ Xamarin: by Microsoft
 - ▶ C#
 - ▶ Flutter: by Google
 - ▶ Dart
 - ▶ React Native: by Facebook
 - ▶ JavaScript

Dev Communities

Cross-platform Mobile Frameworks Used by Software Developers Worldwide in 2019 and 2020



Source: Statista 2021



Introduction to React Native

- ▶ React Native:
 - ▶ JavaScript framework for developing real, native apps for iOS and Android
 - ▶ Based on ReactJS – a framework mainly for web development
 - ▶ Using native components
- ▶ Learning curve

